

ECON3200 Monetary Economics

Final Exam Recap: Lectures 6–11

Lecture-focused notes based on slides, tutorials, recap material, and mock exam style

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1 How to Use These Notes

The final exam focuses on material from Lecture and Tutorial 6 onwards. The exam style is mainly about explaining mechanisms, not only writing formulas.

The main cross-lecture logic is:

nominal rigidities $\Rightarrow$ money affects real variables in the short run
 $\Rightarrow$ NK model explains the mechanism
 $\Rightarrow$ policy design depends on credibility and expectations.

For every shock question, first identify where the shock enters the model. Then explain the movement through the IS curve, Phillips curve, and Taylor rule.

2 Lecture 6 Recap: Money in the Short Run

2.1 What Lecture 6 Focuses On

Lecture 6 explains why money can be non-neutral in the short run. The key reason is price and wage rigidity. Because prices and wages do not adjust immediately, monetary policy can affect real variables such as real wages, real interest rates, output, and employment.

The lecture focuses on:

1. empirical evidence on monetary policy shocks;
2. one-period nominal wage contracts;
3. jobless recovery and downward nominal wage rigidity.

2.2 Empirical Evidence of Monetary Policy Shocks

A contractionary monetary policy shock means the policy interest rate rises unexpectedly:

$$\begin{aligned}\hat{i}_t \uparrow &\Rightarrow \hat{r}_t \uparrow \\ &\Rightarrow \text{demand} \downarrow \\ &\Rightarrow \hat{y}_t \downarrow \\ &\Rightarrow \hat{\pi}_t \downarrow \text{ slowly} \\ &\Rightarrow \hat{m}_t \downarrow .\end{aligned}$$

The fall in output is usually hump-shaped. Output does not necessarily fall by the largest amount immediately. It falls after some delay and then gradually returns to steady state. Inflation or the GDP deflator responds slowly because prices are sticky.

The negative relationship between money supply and the nominal interest rate is called the liquidity effect:

$$\hat{i}_t \uparrow, \quad \hat{m}_t \downarrow .$$

For an expansionary monetary policy shock, the movements are reversed:

$$\begin{aligned}\hat{i}_t \downarrow &\Rightarrow \hat{r}_t \downarrow \\ &\Rightarrow \text{demand} \uparrow \\ &\Rightarrow \hat{y}_t \uparrow \\ &\Rightarrow \hat{\pi}_t \uparrow \text{ slowly} \\ &\Rightarrow \hat{m}_t \uparrow .\end{aligned}$$

2.3 One-Period Nominal Wage Contract

The real wage equals the marginal product of labour:

$$\frac{w_t^N}{P_t} = f_l(k_{t-1}, l_t).$$

With a one-period-ahead nominal wage contract, the nominal wage is set before period t :

$$w_t^N = E_{t-1} [P_t f_l(k_{t-1}, l_t)].$$

Once prices are known in period t , firms choose labour so that

$$P_t f_l(k_{t-1}, l_t) = E_{t-1} [P_t f_l(k_{t-1}, l_t)].$$

Assume production is

$$y_t = A_t k_{t-1}^\alpha l_t^{1-\alpha}, \quad A_t = e^{a_t}, \quad a_t = \rho_a a_{t-1} + \omega_t.$$

The linearised output equation is

$$\hat{y}_t = \alpha \hat{k}_{t-1} + (1 - \alpha) \hat{l}_t + a_t.$$

The linearised nominal marginal product of labour is

$$\hat{P}_t + \alpha \hat{k}_{t-1} - \alpha \hat{l}_t + a_t.$$

If the economy was at steady state in period $t - 1$, $\hat{k}_{t-1} = 0$, and no shocks were expected, then

$$\begin{aligned} \hat{l}_t &= \frac{1}{\alpha} \left(\hat{P}_t - E_{t-1}[\hat{P}_t] + \omega_t \right), \\ \hat{y}_t &= \frac{1 - \alpha}{\alpha} \left(\hat{P}_t - E_{t-1}[\hat{P}_t] \right) + \frac{1}{\alpha} \omega_t. \end{aligned}$$

2.4 Mechanisms

Positive price surprise:

$$\begin{aligned} \hat{P}_t > E_{t-1}[\hat{P}_t] &\Rightarrow \text{real wage falls} \\ &\Rightarrow \hat{l}_t \uparrow \\ &\Rightarrow \hat{y}_t \uparrow. \end{aligned}$$

Negative price surprise:

$$\begin{aligned} \hat{P}_t < E_{t-1}[\hat{P}_t] &\Rightarrow \text{real wage rises} \\ &\Rightarrow \hat{l}_t \downarrow \\ &\Rightarrow \hat{y}_t \downarrow. \end{aligned}$$

Positive productivity shock:

$$\omega_t > 0 \Rightarrow \hat{l}_t \uparrow, \quad \hat{y}_t \uparrow.$$

Negative productivity shock:

$$\omega_t < 0 \Rightarrow \hat{l}_t \downarrow, \quad \hat{y}_t \downarrow.$$

2.5 Jobless Recovery

Downward nominal wage rigidity means nominal wages cannot easily fall:

$$w_t^N \geq (1 + \gamma)w_{t-1}^N.$$

Firms' labour demand condition is

$$\frac{w_t^N}{P_t} = A_t \alpha l_t^{\alpha-1}.$$

If full employment held in period $t-1$, and wage growth was set using expected productivity growth and expected inflation,

$$1 + \gamma = (1 + g_t^e)(1 + \pi_t^e),$$

then unemployment can arise if actual productivity growth or inflation is lower than expected:

$$g_t < g_t^e \quad \text{or} \quad \pi_t < \pi_t^e.$$

Mechanism:

$$\begin{aligned} \text{actual inflation/growth lower than expected} &\Rightarrow \text{nominal wage too high} \\ &\Rightarrow \text{real wage too high} \\ &\Rightarrow \text{labour demand falls} \\ &\Rightarrow l_t < 1. \end{aligned}$$

This explains why output can recover while employment remains weak.

3 Lecture 7 Recap: Basic New Keynesian Model

3.1 What Lecture 7 Focuses On

Lecture 7 builds the basic New Keynesian model. The focus is the short-run transmission mechanism of monetary policy through households, firms, and the central bank.

3.2 The Three-Equation NK Model

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t, \quad (1.NK^*)$$

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t, \quad (2.NK^*)$$

$$\hat{i}_t = \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t. \quad (3.NK^*)$$

The output gap is

$$\hat{x}_t = \hat{y}_t - \hat{y}_t^f.$$

Here $\hat{y}_t$ is actual output and $\hat{y}_t^f$ is flexible-price output.

3.3 Economic Meaning

NK IS curve:

$$\hat{r}_t = \hat{i}_t - E_t[\hat{\pi}_{t+1}].$$

If the real interest rate rises, current demand falls:

$$\hat{r}_t \uparrow \Rightarrow \hat{x}_t \downarrow.$$

If the real interest rate falls, current demand rises:

$$\hat{r}_t \downarrow \Rightarrow \hat{x}_t \uparrow.$$

NK Phillips curve:

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t.$$

If $\hat{x}_t > 0$, marginal costs rise and inflation rises:

$$\hat{x}_t > 0 \Rightarrow \hat{\pi}_t \uparrow.$$

If $\hat{x}_t < 0$, inflation falls:

$$\hat{x}_t < 0 \Rightarrow \hat{\pi}_t \downarrow.$$

Taylor rule:

$$\hat{i}_t = \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t.$$

The central bank raises the nominal interest rate when inflation or the output gap rises.

3.4 Monetary Policy Shock

A monetary policy shock enters through

$$v_t = \rho_v v_{t-1} + \varepsilon_t^v.$$

Positive monetary policy shock: contractionary shock

$$\begin{aligned} \varepsilon_t^v > 0 &\Rightarrow v_t \uparrow \\ &\Rightarrow \hat{i}_t \uparrow \\ &\Rightarrow \hat{r}_t \uparrow \\ &\Rightarrow \hat{x}_t \downarrow \\ &\Rightarrow \hat{\pi}_t \downarrow. \end{aligned}$$

Since technology has not changed, flexible-price output does not move much:

$$\hat{y}_t \downarrow, \quad \hat{y}_t^f \text{ unchanged}, \quad \hat{x}_t \downarrow.$$

Negative monetary policy shock: expansionary shock

$$\begin{aligned} \varepsilon_t^v < 0 &\Rightarrow v_t \downarrow \\ &\Rightarrow \hat{i}_t \downarrow \\ &\Rightarrow \hat{r}_t \downarrow \\ &\Rightarrow \hat{x}_t \uparrow \\ &\Rightarrow \hat{\pi}_t \uparrow. \end{aligned}$$

3.5 Traditional Phillips Curve vs NK Phillips Curve

The NK Phillips curve is forward-looking and derived from an optimising model. The traditional Phillips curve is backward-looking. Both link inflation and the output gap, but the NK output gap is

$$\hat{y}_t - \hat{y}_t^f,$$

not simply output relative to trend.

4 Lecture 8 Recap: Shocks to the NK Economy

4.1 What Lecture 8 Focuses On

Lecture 8 studies how the NK economy responds to technology, spending, and cost-push shocks. The exam focus is to know where each shock enters and how it moves output, inflation, employment, and interest rates.

4.2 Technology Shock

Productivity follows

$$a_t = \rho_a a_{t-1} + \omega_t, \quad \hat{z}_t = a_t.$$

Flexible-price output is

$$\hat{y}_t^f = \left(\frac{1 + \phi}{\sigma + \phi} \right) \hat{z}_t.$$

Positive technology shock:

$$\begin{aligned} \omega_t > 0 &\Rightarrow \hat{z}_t \uparrow \\ &\Rightarrow \hat{y}_t^f \uparrow \text{ more than } \hat{y}_t \\ &\Rightarrow \hat{x}_t = \hat{y}_t - \hat{y}_t^f \downarrow \\ &\Rightarrow \hat{\pi}_t \downarrow \\ &\Rightarrow \hat{i}_t \downarrow. \end{aligned}$$

Employment is

$$\hat{l}_t = \hat{y}_t - \hat{z}_t.$$

Since productivity rises more than actual output,

$$\hat{l}_t \downarrow.$$

Negative technology shock:

$$\begin{aligned} \omega_t < 0 &\Rightarrow \hat{z}_t \downarrow \\ &\Rightarrow \hat{y}_t^f \downarrow \text{ more than } \hat{y}_t \\ &\Rightarrow \hat{x}_t \uparrow \\ &\Rightarrow \hat{\pi}_t \uparrow \\ &\Rightarrow \hat{i}_t \uparrow. \end{aligned}$$

If productivity falls more than actual output,

$$\hat{l}_t \uparrow.$$

4.3 Important Business-Cycle Point

A positive technology shock can make output rise while employment falls. However, in the data, output and employment often move together. Therefore, technology shocks may not be the main driver of normal business-cycle fluctuations. Demand shocks and cost-push shocks may be more relevant.

4.4 Spending / Demand Shock

Spending follows

$$g_t = \rho_g g_{t-1} + \omega_t^g.$$

Flexible-price output becomes

$$\hat{y}_t^f = \left(\frac{1 + \phi}{\sigma + \phi} \right) \hat{z}_t + \left(\frac{1}{\sigma + \phi} \right) g_t.$$

The IS curve becomes

$$\begin{aligned} \hat{x}_t &= E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t + u_t^g, \\ u_t &= E_t[\hat{y}_{t+1}^f] - \hat{y}_t^f, \\ u_t^g &= \frac{1}{\sigma} (g_t - E_t[g_{t+1}]). \end{aligned}$$

Positive spending shock:

$$\begin{aligned} g_t \uparrow &\Rightarrow u_t^g \uparrow \\ &\Rightarrow \hat{x}_t \uparrow \\ &\Rightarrow \hat{\pi}_t \uparrow \\ &\Rightarrow \hat{i}_t \uparrow. \end{aligned}$$

Negative spending shock:

$$\begin{aligned} g_t \downarrow &\Rightarrow u_t^g \downarrow \\ &\Rightarrow \hat{x}_t \downarrow \\ &\Rightarrow \hat{\pi}_t \downarrow \\ &\Rightarrow \hat{i}_t \downarrow. \end{aligned}$$

4.5 Cost-Push Shock

With a cost-push shock, the Phillips curve becomes

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t + \varepsilon_t^w.$$

The cost-push shock follows

$$\varepsilon_t^w = \rho_w \varepsilon_{t-1}^w + \varepsilon_t^o.$$

Positive cost-push shock:

$$\begin{aligned} \varepsilon_t^w \uparrow &\Rightarrow \hat{\pi}_t \uparrow \quad \text{directly} \\ &\Rightarrow \hat{i}_t \uparrow \end{aligned}$$

$$\begin{aligned}\Rightarrow \hat{r}_t &\uparrow \\ \Rightarrow \hat{x}_t &\downarrow.\end{aligned}$$

This creates a trade-off:

$$\text{stabilise inflation} \Rightarrow \hat{x}_t < 0, \quad \text{stabilise output} \Rightarrow \hat{\pi}_t > 0.$$

Negative cost-push shock:

$$\begin{aligned}\varepsilon_t^w \downarrow &\Rightarrow \hat{\pi}_t \downarrow \quad \text{directly} \\ &\Rightarrow \hat{i}_t \downarrow \\ &\Rightarrow \hat{r}_t \downarrow \\ &\Rightarrow \hat{x}_t \uparrow.\end{aligned}$$

5 Lecture 9 Recap: Open Economy, Monetary Policy Rules, and Trade-off

5.1 What Lecture 9 Focuses On

Lecture 9 extends the NK model to a small open economy. Monetary policy works through the usual real-interest-rate channel and an extra exchange-rate or terms-of-trade channel. The lecture also studies stability under different policy rules.

5.2 Terms of Trade and CPI Inflation

The terms of trade are

$$\hat{s}_t = \hat{p}_t^f - \hat{p}_t^h, \quad \Delta \hat{s}_t = \hat{s}_t - \hat{s}_{t-1}.$$

If $\hat{s}_t$ rises, foreign goods become relatively more expensive. If $\hat{s}_t$ falls, foreign goods become relatively cheaper.

The parameter $\gamma \in (0, 1)$ measures openness. CPI inflation is

$$\hat{\pi}_t^c = \hat{\pi}_t^h + \gamma \Delta \hat{s}_t.$$

Therefore,

$$\Delta \hat{s}_t > 0 \Rightarrow \hat{\pi}_t^c \uparrow, \quad \Delta \hat{s}_t < 0 \Rightarrow \hat{\pi}_t^c \downarrow.$$

5.3 Open-Economy NK System

$$\begin{aligned}\hat{s}_t &= (\hat{i}_t^* - E_t[\hat{\pi}_{t+1}^{c*}]) - (\hat{i}_t - E_t[\hat{\pi}_{t+1}^c]) + E_t[\hat{s}_{t+1}], \\ \hat{x}_t &= E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}^c] \right) - \frac{\gamma\psi}{\sigma} E_t[\Delta \hat{s}_{t+1}] + u_t, \\ \hat{\pi}_t^c &= \beta E_t[\hat{\pi}_{t+1}^c] + \tilde{\kappa}_o \hat{x}_t - \beta \gamma E_t[\Delta \hat{s}_{t+1}] + \gamma \Delta \hat{s}_t, \\ \hat{i}_t &= \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t.\end{aligned}$$

5.4 Monetary Policy Shock in an Open Economy

Contractionary shock:

Standard demand channel:

$$\begin{aligned}\hat{i}_t \uparrow &\Rightarrow \hat{r}_t \uparrow \\ &\Rightarrow \hat{x}_t \downarrow \\ &\Rightarrow \hat{\pi}_t^c \downarrow.\end{aligned}$$

Exchange-rate channel:

$$\begin{aligned}\hat{i}_t \uparrow &\Rightarrow \text{domestic currency appreciates} \\ &\Rightarrow \hat{s}_t \downarrow \\ &\Rightarrow \Delta \hat{s}_t < 0 \\ &\Rightarrow \hat{\pi}_t^c \downarrow.\end{aligned}$$

Expansionary shock:

Standard demand channel:

$$\begin{aligned}\hat{i}_t \downarrow &\Rightarrow \hat{r}_t \downarrow \\ &\Rightarrow \hat{x}_t \uparrow \\ &\Rightarrow \hat{\pi}_t^c \uparrow.\end{aligned}$$

Exchange-rate channel:

$$\begin{aligned}\hat{i}_t \downarrow &\Rightarrow \text{domestic currency depreciates} \\ &\Rightarrow \hat{s}_t \uparrow \\ &\Rightarrow \Delta \hat{s}_t > 0 \\ &\Rightarrow \hat{\pi}_t^c \uparrow.\end{aligned}$$

Therefore, monetary policy has a stronger effect in a small open economy than in a closed economy. The effect is stronger when γ is larger.

5.5 Monetary Policy Rules and Stability

Rule 3A: Interest-rate peg

$$\hat{i}_t = \rho_i \hat{i}_{t-1} + v_t.$$

If expected inflation rises but the nominal interest rate does not respond,

$$\begin{aligned}E_t[\hat{\pi}_{t+1}] \uparrow &\Rightarrow \hat{r}_t \downarrow \\ &\Rightarrow \hat{x}_t \uparrow \\ &\Rightarrow \hat{\pi}_t \uparrow.\end{aligned}$$

This can create self-fulfilling expectations and multiple equilibria.

Rule 3B: inflation-only Taylor rule

$$\hat{i}_t = \delta \hat{\pi}_t + v_t.$$

The economy has a unique equilibrium if

$$\delta > 1.$$

This is the Taylor Principle.

Rule 3C: inflation and output-gap Taylor rule

$$\hat{i}_t = \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t.$$

The stability condition is

$$\tilde{\kappa}(\delta_\pi - 1) + (1 - \beta)\delta_x > 0.$$

5.6 Trade-off in Monetary Policy Design

Without cost-push shocks, the central bank can stabilise both inflation and the output gap:

$$\hat{x}_t = 0, \quad \hat{\pi}_t = 0.$$

With a cost-push shock,

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t + \varepsilon_t^w.$$

Solving forward gives

$$\hat{\pi}_t = \tilde{\kappa} \sum_{k=0}^{\infty} \beta^k E_t[\hat{x}_{t+k}] + \varepsilon_t^w.$$

If $\varepsilon_t^w > 0$, inflation rises directly. To keep inflation at target, the central bank needs a negative current or expected future output gap:

$$\hat{x}_{t+k} < 0.$$

Therefore,

$$\text{stabilise inflation} \Rightarrow \text{lower output gap}, \quad \text{stabilise output} \Rightarrow \text{higher inflation}.$$

6 Lecture 10 Recap: Time-Inconsistency Problem

6.1 What Lecture 10 Focuses On

Lecture 10 explains why discretionary monetary policy can generate inflation bias. Surprise inflation may raise output temporarily, but rational agents anticipate the central bank's incentive to inflate. The result is higher expected inflation without a permanent output gain.

6.2 Barro–Gordon Model

The Phillips curve is

$$y = y^n + a(\pi - \pi^e) + e.$$

If actual inflation is higher than expected,

$$\pi > \pi^e \Rightarrow y > y^n.$$

If actual inflation equals expected inflation,

$$\pi = \pi^e \Rightarrow y = y^n + e.$$

The policy instrument can be written as

$$\pi = \zeta + v.$$

6.3 Discretion

Under discretion, the central bank chooses policy after expectations are formed. It is tempted to create surprise inflation to push output above natural output.

With rational expectations, people anticipate this. Expected inflation under discretion is

$$\pi^e = \lambda a k.$$

If the lecture normalises $k = 1$, this becomes

$$\pi^e = \lambda a.$$

Expected output is still natural output:

$$E[y] = y^n.$$

Therefore,

$$\text{discretion} \Rightarrow \pi^e > 0, \quad E[y] = y^n.$$

This is inflation bias: higher inflation without higher expected output.

6.4 Commitment

Under commitment, the central bank can credibly commit to low inflation:

$$\pi^e = 0, \quad E[y] = y^n.$$

Commitment avoids inflation bias and gives higher expected welfare than discretion.

6.5 Reputation

Reputation can make low inflation credible. If the central bank behaves, the public expects low inflation. If it cheats, reputation is lost and the public expects high inflation next period.

Trigger strategy:

$$\begin{aligned} \pi_t^e &= \bar{\pi} < \lambda a && \text{if the central bank behaved last period,} \\ \pi_t^e &= \lambda a && \text{otherwise.} \end{aligned}$$

The gain from deviating is

$$G(\bar{\pi}) = \frac{1}{2}(\lambda a - \bar{\pi})^2.$$

The cost of deviating is

$$C(\bar{\pi}) = \frac{1}{2}\beta [(\lambda a)^2 - \bar{\pi}^2].$$

A low inflation rate can be supported by reputation if

$$C(\bar{\pi}) \geq G(\bar{\pi}).$$

7 Lecture 11 Recap: Solutions to Inflation Bias

7.1 What Lecture 11 Focuses On

Lecture 11 studies institutional solutions to inflation bias. The common theme is commitment. Since a discretionary central bank is tempted to create surprise inflation, society can design institutions or incentives that make low inflation more credible.

Main solutions:

1. reputation;
2. delegation to a conservative central bank;
3. contracts with the central banker;
4. central bank independence;
5. targeting rules.

7.2 Delegation to a Conservative Central Bank

A conservative central banker places greater weight on inflation fighting:

$$E[V] = E \left[\frac{1}{2} \lambda (y - y^n - k)^2 + \frac{1}{2} (1 + \delta) \pi^2 \right].$$

Here $\delta > 0$ measures conservatism. Expected inflation becomes

$$\pi^e(\delta) = \frac{\lambda a k}{1 + \delta}.$$

Thus,

$$\delta \uparrow \Rightarrow \pi^e(\delta) \downarrow.$$

The cost is weaker output stabilisation:

$$\delta \uparrow \Rightarrow \text{inflation bias lower but output stabilisation weaker.}$$

7.3 Central Bank Independence

Central bank independence can act as a commitment device. It makes low-inflation policy more credible by reducing political pressure to create surprise inflation.

The limitation is that delegation itself may face a commitment problem. Once expectations are set, the government may want to replace a conservative central banker with a more flexible one.

7.4 Contracts with the Central Banker

Walsh's solution is to give the central banker a contract that penalises inflation. Let the transfer be $t(\pi)$, where

$$t'(\pi) < 0.$$

The central banker's utility is

$$U = t(\pi) - V.$$

Expected inflation becomes

$$\pi^e = \lambda a k + E[t'].$$

The optimal transfer is

$$t = t_0 - \lambda a k \pi.$$

Then

$$E[t'] = -\lambda a k, \quad \pi^e = 0.$$

This eliminates inflation bias while preserving flexibility to respond to shocks. The limitation is that the government must commit to enforcing the contract.

7.5 Targeting Rules

A flexible targeting rule adds a penalty for deviations from an inflation target:

$$\begin{aligned} & \frac{1}{2} \lambda E_t [y - y^n - k]^2 + \frac{1}{2} E_t [\pi_t - \pi^*]^2 \\ & + \frac{1}{2} h E_t [\pi_t - \pi^T]^2. \end{aligned}$$

Expected inflation becomes

$$\pi^e = \frac{1}{1+h} [k \lambda a + \pi^* + h \pi^T].$$

If $\pi^T = \pi^* = 0$, then

$$\pi^e = \frac{1}{1+h} k \lambda a.$$

So flexible inflation targeting reduces inflation bias, but does not necessarily eliminate it unless the penalty is very strong.

Strict targeting can be written as

$$\zeta = \pi^*.$$

Then

$$\pi^e = \pi^*.$$

If $\pi^* = 0$, inflation bias is eliminated, but flexibility is reduced.

8 Final Cross-Lecture Shock Summary

Shock / Topic	Positive shock	Negative shock
Price surprise	$\hat{P}_t > E_{t-1}\hat{P}_t$: real wage falls; labour and output rise.	$\hat{P}_t < E_{t-1}\hat{P}_t$: real wage rises; labour and output fall.
Productivity in wage-contract model	$\omega_t > 0$: productivity, labour, and output rise.	$\omega_t < 0$: productivity, labour, and output fall.
Monetary policy in NK model	$v_t > 0$: contractionary; $i \uparrow, r \uparrow, x \downarrow, \pi \downarrow, y \downarrow$.	$v_t < 0$: expansionary; $i \downarrow, r \downarrow, x \uparrow, \pi \uparrow, y \uparrow$.
Technology in NK model	$z \uparrow, y^f \uparrow$ more than y ; so $x \downarrow, \pi \downarrow, i \downarrow, l \downarrow$.	$z \downarrow, y^f \downarrow$ more than y ; so $x \uparrow, \pi \uparrow, i \uparrow, l \uparrow$.
Spending / demand	Demand rises; $x \uparrow, \pi \uparrow, i \uparrow$.	Demand falls; $x \downarrow, \pi \downarrow, i \downarrow$.
Cost-push	Direct $\pi \uparrow$; central bank raises i , so $x \downarrow$. Trade-off.	Direct $\pi \downarrow$; central bank lowers i , so $x \uparrow$.
Terms of trade	$\Delta s_t > 0$: foreign goods more expensive; CPI inflation rises.	$\Delta s_t < 0$: foreign goods cheaper; CPI inflation falls.
Open-economy MP	Contractionary: $i \uparrow \Rightarrow x \downarrow$, appreciation, $s \downarrow, \pi^c \downarrow$.	Expansionary: $i \downarrow \Rightarrow x \uparrow$, depreciation, $s \uparrow, \pi^c \uparrow$.
Barro–Gordon surprise inflation	$\pi > \pi^e$: output temporarily above natural output.	$\pi < \pi^e$: output temporarily below natural output.

9 Final Exam-Style Templates

9.1 Monetary Policy Evidence

A contractionary monetary policy shock raises interest rates, lowers output with a hump-shaped response, lowers inflation only gradually, and lowers money supply. An expansionary shock reverses these movements.

9.2 NK vs Traditional Phillips Curve

The NK Phillips curve is forward-looking and derived from an optimising model. The traditional Phillips curve is backward-looking. Both link inflation and the output gap, but the NK output gap is

$$\hat{y}_t - \hat{y}_t^f.$$

9.3 Taylor Principle

A Taylor rule targeting only inflation can lead to multiple equilibria if the Taylor Principle is not satisfied. The Taylor Principle requires

$$\delta > 1.$$

If it is satisfied, the real interest rate rises when inflation rises, which stabilises the economy.

9.4 Technology Shock

A positive technology shock increases output, but flexible-price output increases more. Therefore, the output gap falls, inflation falls, interest rates fall, and employment falls.

9.5 Cost-Push Shock and Trade-off

A positive cost-push shock raises inflation directly. To reduce inflation, the central bank must raise interest rates and create a negative output gap. Hence, cost-push shocks create a trade-off between stabilising inflation and stabilising output.

9.6 Inflation Bias

Inflation bias arises because the central bank cannot commit to low inflation and is tempted to create surprise inflation. Under rational expectations, people anticipate this, so there is positive expected inflation but no expected output gain.

9.7 Solutions to Inflation Bias

Reputation, delegation, contracts, central bank independence, and targeting rules reduce inflation bias by changing expectations or incentives. The main cost is usually less flexibility or another commitment problem.

10 One-Sentence Takeaway

The second half of ECON3200 explains that monetary policy matters in the short run because nominal rigidities affect real interest rates and output. However, effective monetary policy also requires credible institutions because discretionary policymakers are tempted to create surprise inflation.